

Seating & Event Availability Tracking System (SEATS) for La Loy Restaurant

Milestone 2 · Part 1 GenAI Log

2110722 Software Project Management

Submitted to

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Members

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GenAI Log

This log records the prompts with which the members produced the Milestone 2 artifacts of the Seating & Event Availability Tracking System (SEATS) project for La Loy Restaurant, and how each initial output was evaluated and revised. Each member submits two prompt log records, one per page. The prompts of records 1 to 6 are written in the CREATE framework (Character, Request, Examples, Adjustments, Type of output, Evaluation), those of records 7 and 8 in the RTF framework (Role, Task, Format) because they were quick requests to an agent that already had the project files as context, and every output was evaluated against the three course criteria: requirement satisfaction, PMBOK correctness and project consistency, with the Milestone 1 report and with the other Milestone 2 artifacts. GenAI is used as an assistant rather than a project manager: every output is reviewed and, where needed, revised by the member, who remains responsible for the artifact.

All prompts draw on one single source of truth, the Milestone 1 Part 2 Final Deliverables (project dates, release scope, role rates, phase effort totals and budget), so that the independently generated artifacts stay mutually consistent; where the AI's output was a structure or a draft, the figures of the final tables were computed from the project's plan model, so that the WBS, effort, assignment, cost and schedule tables cannot contradict each other. Members prompted through their own tool: Boonnisa Tangjitpermkwamdee used ChatGPT (drafting assistance); Sirapatr Chalermwattanatrai used Google Gemini (gemini.google.com); Chawanakorn Sanannam used the OpenCode terminal CLI (opencode.ai) with the model "Big Pickle" (opencode/big-pickle); Watayut Aiamanan used Claude Code (Anthropic, model Claude Fable 5.1) inside the project repository, so that the model read the Milestone 1 sources, the drafts and the financial model directly and wrote its results into the report sources. Every record is linked to its row of the Review Board Decision Log (Part 2, Table 1.1).

Table 0.1 Prompt log records of Milestone 2

Record	Artifact	Member (prompt design)	RUP role of the member and why the material is theirs	Artifact owner (RUP)	Framework
1	Scope statement and deliverables	Boonnisa T.	Project Manager and Change Control Manager: owns the scope baseline and the scope statement, so the scope wording and the deliverable categories are her responsibility	Project Manager	CREATE
2	Success criteria and budget baseline	Boonnisa T.	Project Manager and Change Control Manager: accountable for the cost baseline and the measurable success criteria agreed with the sponsor	Project Manager	CREATE
3	WBS and detailed schedule	Sirapatr C.	System Analyst and Software Architect: owns the decomposition of the requirements and architecture into work packages and the derivation of the task list from the RUP iterations	System Analyst / Software Architect	CREATE
4	Milestones and critical path	Sirapatr C.	System Analyst and Software Architect: derives the logical dependency chain and the critical path from the technical work packages and the RUP milestones	System Analyst / Software Architect	CREATE
5	Effort estimation for tasks	Chawanakorn S.	Requirements Specifier and Implementer: estimates the hours of the feature work he will build, so the estimates come from the person doing the work	Requirements Specifier	CREATE
6	WBS decomposition and features by release	Chawanakorn S.	Requirements Specifier and User-Interface Designer: turns the feature set into the deliverable-oriented decomposition and keeps each feature in exactly one release	Requirements Specifier	CREATE

Record	Artifact	Member (prompt design)	RUP role of the member and why the material is theirs	Artifact owner (RUP)	Framework
7	Consistency review of the Milestone 2 drafts, quality requirements and verification	Watayut A.	Quality Assurance and Tester: reviews every artifact against the criteria and the single source of truth, and maps every quality target to an independent test	Quality Assurance / Test Designer	RTF
8	Cost estimate consistency and the plan model	Watayut A.	Quality Assurance and Tester: independently verifies that effort, assignment, cost and schedule agree with each other and with Milestone 1	Quality Assurance / Test Designer	RTF

Record 1: Scope statement and deliverables

1. Artifact	Project scope statement (Part 2, Section 2): project summary, product characteristics and requirements, project-management and product-related deliverables, exclusions and success criteria for Release 1.0 Core, Release 2.0 Growth and Release 2.1 productization.
2. Member	Boonnisa Tangjitpermkwamdee
3. AI Tool	ChatGPT (drafting assistance). The Milestone 1 report was supplied as the single source of truth.
4. Initial Prompt	**[character]** You are a project manager who owns the scope baseline. **[request]** Using the SEATS report for La Loy Restaurant, draft a scope statement with project summary, product characteristics, project-management and product deliverables, exclusions and measurable success criteria. **[examples]** Separate Release 1.0 Core, Release 2.0 Growth and Release 2.1 productization. **[adjustments]** Do not invent requirements; keep the release boundaries of the source report. **[type of output]** A scope statement table with a clear deliverable category column. **[evaluation]** Every deliverable category must be distinct and the exclusions must be explicit.
5. Initial Output	The initial draft summarized booking, payment, QR check-in and staff functions, but did not clearly separate project-management, product and closure deliverables, and the exclusions were missing.
6. Review	Review Board row 1: the draft mixed product deliverables with project-management deliverables and did not clearly list exclusions. Requirement satisfaction was partial; PMBOK correctness required a single scope statement with explicit exclusions; project consistency required the release boundaries of the source report to be retained.
7. Revision	Asked for separate project-management and product deliverables, explicit exclusions and measurable, source-based success criteria. By hand, the product characteristics were restated with the feature identifiers F1 to F37 of Milestone 1 instead of the requirement labels the AI had invented, the closure deliverables were folded into the project-management deliverables as the brief's two categories ask, and the exclusions were tied to the Future features and the restaurant-paid items of Milestone 1.
8. Final Output	Part 2, Sections 2.1, 2.3 and 2.4, as delivered below (the feature table and the quality table are in Record 7 and Part 2, Section 2.2; the success criteria in Record 2). Acceptable because the deliverables are separated into the two categories of the brief, the exclusions are explicit and the release boundaries and dates are those of Milestone 1. Review Board decision: Revise; accepted after the revision. **2.1 Project Summary** SEATS is a LINE-integrated web application for reserving tables at La Loy Restaurant's live-music concerts. Customers use a LIFF web application opened from the Rich Menu of the restaurant's LINE

Official Account to browse concert rounds, view a live table map, select a table, hold it for 15 minutes, pay the full table fee through a payment gateway and receive a QR e-ticket. Restaurant staff use a shared back-office to manage rounds, layouts and prices, watch the live booking view, handle exceptions and scan QR codes at the door. One central record replaces the chat, the hand-edited image and the notebook, so that double bookings and payment disputes disappear and the Manager handles exceptions only. After the build, the system is run through a pilot year at La Loy and productized as a multi-tenant subscription service for other live-music restaurants.

The project runs from 3 August 2026 to 28 July 2028 (two project years) under the tailored Rational Unified Process of the Development Case: Year 0 builds and deploys Release 1.0 Core (in operation 2 April 2027) and Release 2.0 Growth (30 July 2027); Year 1 runs the live pilot and delivers Release 2.1 Product (11 February 2028), and closes with the expansion decision of the software house on 28 July 2028. The project budget is 1,010,590 THB (charter, Table 4.1 of Milestone 1), of which La Loy's pilot implementation fee covers 150,000 THB. The scope statement, the WBS of Section 3 and the WBS dictionary of Section 3.3 form the scope baseline; changes to it go through the Change Control Board.

Table 2.3 Project deliverables

Category	Deliverables
Project-management deliverables	Business case with financial analysis, project charter, stakeholder list and project organization (Milestone 1); scope statement, WBS and WBS dictionary, effort, resource and cost estimates, schedule and cost baseline (this document); quality, risk and communication management plans (Milestone 3); Development Case and Software Development Plan; iteration plan and assessment of every iteration; weekly status assessments; change-request decisions of the Change Control Board; Risk List updates; milestone reviews (LCO, LCA, IOC, Product Release 1.0, 2.0 and 2.1); monthly pilot KPI reports; pilot review, expansion decision record, hand-over record, lessons learned and sign-off
Product-related deliverables	Requirements and design artifacts of the tailored RUP: Vision, Glossary, Stakeholder Requests, Business Rules, Domain Model, Use-Case Model and use-case specifications, Software Requirements Specification, Supplementary Specification, Analysis Model, Software Architecture Document, user-interface prototype, test plan and test cases; the tested increment of every Construction iteration; Release 1.0 Core in operation at La Loy (2 April 2027), Release 2.0 Growth (30 July 2027) and the productized Release 2.1 (11 February 2028); load-test, system-test and acceptance records; training and user guide for the Manager and the Front Staff; migration of the current bookings and, in Release 2.1, of La Loy to the productized service; production environment, support routine and Free-plan sign-up

2.4 Exclusions

- Split payment among the members of a party (F36) and table photos with distance to the stage (F37): Future candidates outside the project (Table 2.13 of Milestone 1).
- Restaurant-configurable booking rules beyond the rules agreed for the defined releases (hold, grace, transfer cut-off, waitlist, refund policy are fixed by the Owner's decisions before LCA).
- Payment-gateway fees, each restaurant's LINE Official Account plan and the pilot restaurant's own check-in tablet (about 9,000 THB): paid by the restaurant (CON-03, CON-07), not by the project budget.
- Sales, onboarding and operation of the subscription service for other restaurants after the expansion decision (Years 2 to 5 of the financial analysis): outside the project.
- Hardware at the venue beyond the test devices of the software house; any change to the restaurant's own LINE Official Account content.

Record 2: Success criteria and budget baseline

1. Artifact	Success criteria and budget baseline (Part 2, Sections 2.5 and 6): measurable targets with evidence, and the cost baseline (Year 0 767,690 THB; Year 1 242,900 THB; total 1,010,590 THB; labor 846,600 THB; contingency 88,490 THB; net investment 860,590 THB).
2. Member	Boonnisa Tangjitpermkwamdee
3. AI Tool	ChatGPT (drafting assistance). The budget and success-criteria figures of Milestone 1 were supplied as the single source of truth.
4. Initial Prompt	**[character]** You are a project manager checking measurable success criteria and a budget baseline. **[request]** Check the SEATS success criteria and budget against the project period 3 Aug 2026 to 28 Jul 2028, total budget 1,010,590 THB, labor 846,600 THB, pilot fee 150,000 THB and net investment 860,590 THB. **[examples]** Use source targets and flag values needing human verification. **[adjustments]** Include numerical thresholds and evidence for each criterion. **[type of output]** A criteria/evidence table plus a budget reconciliation summary. **[evaluation]** Preserve the reported baseline figures and show how each target is verified.
5. Initial Output	The draft included general success criteria and the headline budget but did not explain how the targets would be verified or distinguish the source figures from assumptions.
6. Review	Review Board row 2: success measures needed numeric targets and evidence linked to the source report; the budget needed a transparent category split. PMBOK correctness: the cost baseline must reconcile.
7. Revision	Requested a criteria/evidence table and a clear distinction between source figures and assumptions, keeping the budget aligned with the source. By hand, the criteria were matched one-to-one with the seven criteria of the charter and the budget summary was replaced by the cost build-up from the work packages (Section 6), which reproduces the figures.
8. Final Output	Part 2, Table 2.4, as delivered below, and the cost reconciliation of Section 6.4 (Record 8). The criteria include booking within 3 minutes, a 35 % no-show reduction, at least 5 transactions a second, response within 2 seconds, check-in within 15 seconds per table and recovery within 2 hours, each with its evidence; the budget figures are retained and reproduced by the bottom-up cost baseline. Review Board decision: Accept. **Table 2.4 Success criteria and evidence**

No.	Success criterion (charter)	Target / evidence
1	Release milestones within budget	Release 1.0 Core in operation at La Loy on 2 April 2027, Release 2.0 Growth on 30 July 2027, Release 2.1 on 11 February 2028; cost at close-out within the project budget of 1,010,590 THB (go-live records, iteration assessments, cost baseline of Section 6)
2	Booking integrity	Zero double bookings on La Loy's concert nights after go-live; table status real-time on one record (audit trail and pilot KPI reports)
3	Customer experience	A booking completed within 3 minutes from table selection to confirmation, without a Manager reply (timestamps of the booking record; sample of pilot nights)
4	Business outcome	No-show rate at La Loy reduced by 35 % within six months of Release 2.0, against the baseline of Table 2.2 of Milestone 1 (check-in records versus bookings)
5	Operational quality and cost	Check-in within 15 seconds per table; at least 5 transactions a second at booking-open time; response within 2 seconds; recovery within 2 hours; the restaurant's hosting, APIs and LINE plan at or below 1,000 THB a month (Table 2.2 tests; monthly cost review)
6	Requirements and acceptance	Vision approved by the Owner at LCO, SRS signed off at LCA, acceptance tests of the two main use cases (reserve a concert table, check in with e-ticket) passed with the Manager and the Front Staff (milestone review and acceptance records)
7	Expansion gate	At the expansion decision: La Loy continues on a paid plan, at least 10 restaurants are on the waiting list, and the per-shop hosting and support cost of the productized service is at or below the assumptions of Table 3.3 of Milestone 1 (pilot review, waiting list, cost report)

Record 3: WBS and detailed schedule

1. Artifact	Work breakdown structure and activity list (Part 2, Sections 3.1 and 3.2): the WBS decomposed to work packages as deliverable-oriented noun phrases, mapped to the ten tailored RUP iterations I1 to T3, and the activity list with WBS ID, activity, duration, predecessors, start, finish and resources.
2. Member	Sirapatr Chalermwattanatrai
3. AI Tool	Google Gemini (gemini.google.com). The Milestone 1 baseline (requirements, the ten RUP iterations and the WBS workstreams) was pasted as the single source of truth.
4. Initial Prompt	**[character]** You are a system analyst building a PMBOK-compliant WBS. **[request]** Make a workstream WBS for SEATS decomposed down to two levels, then convert it into a detailed schedule table with WBS ID, Task, Responsible Role, Duration, Predecessor(s), Start, Finish and Expected Output/Milestone Measure. **[examples]** Align the iteration dates with the SEATS M1 baseline I1 to T3. **[adjustments]** Make every task label a 100 % deliverable-oriented noun phrase rather than a verb action, and label estimated task-level durations as estimates. **[type of output]** A workstream outline plus a schedule-style WBS table. **[evaluation]** Every task must be a noun phrase with an owner, dates, dependencies and an output measure, and the source iteration dates must be preserved.
5. Initial Output	The first WBS was phase-based but several task labels were verb actions (for example "Build table map") and the schedule fields and dependencies were missing.
6. Review	Review Board row 3: the WBS was not 100 % deliverable-oriented and lacked schedule-table columns and clear task dependencies. PMBOK correctness: a WBS needs noun-phrase work packages and a schedule format needs duration, predecessor and output columns; project consistency required the source iteration dates to be preserved.
7. Revision	Re-decomposed the workstreams to two levels with noun-phrase work packages and added the schedule columns (WBS ID, Task, Responsible Role, Duration, Predecessors, Start, Finish, Expected Output), preserving the source dates. By hand, the work packages were taken one level further (49 work packages in eight branches, the features F1 to F35 placed each in exactly one package) and the dates, predecessors and resource units of every activity were computed in the project's plan model inside the iteration windows, with the year-end break of Milestone 1, because the AI's dates had ignored it.
8. Final Output	Part 2, Table 3.1 (the WBS with the activity, dates, dependencies and resources of every work package, keeping the iteration dates of Milestone 1), shown below as the WBS outline with the accountable engineer and the deliverable of each work package. Review Board decision: Revise; accepted after the revision.

WBS outline (the activity, dates, predecessors and resources of every work package are in Part 2, Table 3.1)

WBS ID	Work package	Iteration	Accountable	Deliverable (expected output)
1	**Project Management**			*project management (runs through all phases)*
1.1	Project plans and baselines	I1, I2	Boonnisa T.	Business Case, charter, organization chart, SDP (draft), iteration plans
1.2	Risk List and Development Case	I1, I2	Watayut A.	Development Case v1, Risk List
1.3	Iteration assessments and change control, Elaboration	E1, E2	Boonnisa T.	Status assessments, iteration assessments, change-request decisions
1.4	Iteration assessments and change control, Release 1.0 Construction	C1, C2	Boonnisa T.	Status assessments, iteration assessments, change-request decisions, Risk List updates
1.5	Iteration assessments and change control, Release 1.0 Transition	T1	Boonnisa T.	Status assessments, go-live record
1.6	Iteration assessments and change control, Release 2.0	C3, T2	Boonnisa T.	Status assessments, iteration assessments, change-request decisions
1.7	Iteration assessments and change control, Release 2.1	C4, T3	Boonnisa T.	Status assessments, iteration assessments, change-request decisions
1.8	Pilot KPI reports and reviews	P1	Boonnisa T.	Monthly pilot KPI reports, waiting list of restaurants
1.9	Pilot review, expansion decision and close-out	CL	Boonnisa T.	Pilot review, expansion decision record, hand-over record, lessons learned, sign-off
2	**Inception (I1, I2)**			*Vision, use-case model, LCO*
2.1	Stakeholder requests and problem analysis	I1	Sirapatr C.	Stakeholder Requests, problem and root-cause tables (workshop of RISK-01)
2.2	Vision and Glossary	I1, I2	Sirapatr C.	Vision v1, Glossary
2.3	Use-case model and release plan	I2	Chawanakorn S.	Use-Case Model (outline), release plan
2.4	Review records and LCO package	I2	Watayut A.	Review Records, LCO decision
3	**Elaboration (E1, E2)**			*requirements, architecture, prototype, test plan, LCA*
3.1	Use-case specifications	E1	Chawanakorn S.	Use-case specifications
3.2	Software Requirements Specification and Supplementary Specification	E1, E2	Chawanakorn S.	SRS, Supplementary Specification
3.3	Analysis model and Software Architecture Document	E2	Sirapatr C.	Analysis Model, SAD
3.4	User-interface prototype	E2	Chawanakorn S.	UI prototype, trial records
3.5	Payment gateway and LINE sandbox	E2	Sirapatr C.	Sandbox accounts, onboarding checks, feasibility record
3.6	Test plan and test cases	E2	Watayut A.	Test plan, test cases, load-test design

WBS ID	Work package	Iteration	Accountable	Deliverable (expected output)
3.7	Requirements reviews, SRS baseline and LCA package	E2	Watayut A.	Review Records, SRS baseline, LCA decision
3.8	Scope baseline, estimates, schedule and cost baseline	E2	Boonnisa T.	Scope statement, WBS, estimates, schedule and cost baseline, Construction plan
4	**Release 1.0 Core: Construction (C1, C2)**			*features F1 to F28, IOC*
4.1	Concert rounds, zones and packages	C1	Chawanakorn S.	Back-office round and layout functions, unit tests
4.2	Table map, selection and hold	C1	Sirapatr C.	Customer table map and hold, first-lock-wins locking
4.3	Gateway payment and callback	C1	Sirapatr C.	Payment flow with automatic confirmation
4.4	LINE sign-in, staff accounts and role access	C1	Chawanakorn S.	Sign-in and staff accounts
4.5	C1 integration and iteration test	C1	Watayut A.	Tested C1 increment, iteration assessment
4.6	QR e-ticket and transactional messages	C2	Chawanakorn S.	E-ticket and messages
4.7	Door check-in and name lookup	C2	Chawanakorn S.	Front Staff check-in functions
4.8	Live booking view and exception handling	C2	Sirapatr C.	Manager's live view and exception functions
4.9	Audit trail, backup, consent and push budget	C2	Chawanakorn S.	System functions of Release 1.0
4.10	Load test, system test and IOC package	C2	Watayut A.	Load-test and system-test records, IOC decision
5	**Release 1.0 Core: Transition (T1)**			*acceptance, training, migration, go-live*
5.1	Acceptance tests of Release 1.0	T1	Watayut A.	Acceptance records
5.2	Training and user guide	T1	Chawanakorn S.	Training records, user guide
5.3	Production set-up and data migration	T1	Sirapatr C.	Production system, migrated data
5.4	Pilot concert nights and go-live of Release 1.0	T1	Boonnisa T.	Go-live record, Release 1.0 in operation
6	**Release 2.0 Growth (C3, T2)**			*features F29 to F35, go-live*
6.1	Scheduled reminders	C3	Chawanakorn S.	Reminder functions
6.2	Grace extension and waitlist	C3	Sirapatr C.	Grace and waitlist functions
6.3	Booking transfer	C3	Chawanakorn S.	Transfer function
6.4	Reports and Excel export	C3	Chawanakorn S.	Reports
6.5	C3 integration and system test	C3	Watayut A.	Tested Release 2.0, iteration assessment

WBS ID	Work package	Iteration	Accountable	Deliverable (expected output)
6.6	Acceptance tests and training on Release 2.0	T2	Watayut A.	Acceptance and training records
6.7	Release 2.0 deployment and first concert nights	T2	Boonnisa T.	Go-live record, Release 2.0 in operation, pilot year begins
7	**Pilot operation (P1)**			*concert-night support*
7.1	Concert-night support and maintenance	P1	Watayut A.	Support log, maintenance releases
8	**Release 2.1 Product (C4, T3)**			*productization, migration of La Loy*
8.1	Multi-tenant data model and tenant isolation	C4	Sirapatr C.	Multi-tenant core
8.2	Plans, limits and tier enforcement	C4	Chawanakorn S.	Plan and limit functions
8.3	Subscription billing and company set-up	C4	Chawanakorn S.	Billing functions, company registration, legal documents
8.4	Self-service onboarding and service administration	C4	Chawanakorn S.	Onboarding and administration functions
8.5	Security, load and integration test of Release 2.1	C4	Watayut A.	Test records, iteration assessment
8.6	Migration of La Loy and release of the Free-plan sign-up	T3	Sirapatr C.	La Loy on the productized service, Release 2.1 in operation

Record 4: Milestones and critical path

1. Artifact	Schedule, milestone log and critical path (Part 2, Section 7): the Gantt chart of the project from August 2026 to July 2028 aligned to the ten RUP iterations, the milestone log (LCO, LCA, IOC, Release 1.0, Release 2.0, Release 2.1, expansion decision of 28 July 2028) and the float-based critical-path analysis (TS = LS − ES = LF − EF).
2. Member	Sirapatr Chalermwattanatrai
3. AI Tool	Google Gemini (gemini.google.com). The iteration dates and milestone gates of Milestone 1 were pasted as the single source of truth.
4. Initial Prompt	**[character]** You are a scheduling analyst using formal dependency logic. **[request]** Construct a high-level Gantt chart text view and milestone log from August 2026 to July 2028 that aligns all ten RUP iterations and marks LCO, LCA, IOC, REL 1.0, REL 2.0, REL 2.1 and the 28 July 2028 Expansion Decision Gate, then derive the critical path with TS = LS − ES. **[examples]** Show pilot operation as parallel work with non-zero float. **[adjustments]** Qualify the critical path as a planning analysis rather than a source-approved critical path. **[type of output]** A milestone log, a Gantt chart text view and a float-based critical-path table. **[evaluation]** Source milestones must be unchanged and the critical path must be justified by the float calculation.
5. Initial Output	The draft aligned the iterations to the Gantt view but stated a critical path too definitively and omitted the parallel pilot work with non-zero float.
6. Review	Review Board row 4: Milestone 1 gives iteration dates but does not formally calculate a task-level critical path. PMBOK correctness: the chain must be qualified as a planning analysis with an explicit float formula; project consistency required the source milestone dates to be retained. A second review of the AI's float table found that its early-start dates ignored the year-end break and the gaps between iterations (C2 on 21 December instead of 4 January, T1 on 15 February instead of 1 March), and that the AI had once named the productization branch and once the pilot year as critical.
7. Revision	Applied the standard float formulas (TS = LS − ES = LF − EF), traced the continuous zero-slack path, marked the chain as a planning analysis and showed the parallel branch with its float. By hand, the forward and backward pass were recomputed in the plan model on a working-day calendar with the year-end break, with the close-out linked finish-to-finish to the pilot year, which settles the branch question: the pilot year is critical and the productization branch C4, T3 has 115 working days of total slack; the Gantt chart was drawn from the activity list rather than from the iteration view.

8. Final Output

Part 2, Tables 7.1 and 7.2, as delivered below (the Gantt chart, Figure 7.1, and the discussion of Section 7.2 are in Part 2). Source milestones remain unchanged; the critical path I1 to T2, then the pilot year and the close-out, is justified by the float table and presented as a planning analysis. Review Board decision: Accept.

Table 7.1 Critical path method on the iteration network (working days from 3 August 2026)

ID	Iteration	Duration (wd)	Predecessors	ES	EF	LS	LF	Total slack (wd)	Critical	Early start	Early finish
I1	Inception I1	16	—	0	16	0	16	0	Yes	3 Aug 2026	24 Aug 2026
I2	Inception I2	7	I1	16	23	16	23	0	Yes	25 Aug 2026	2 Sep 2026
E1	Elaboration E1	10	I2	23	33	23	33	0	Yes	3 Sep 2026	16 Sep 2026
E2	Elaboration E2	27	E1	33	60	33	60	0	Yes	17 Sep 2026	23 Oct 2026
C1	Construction C1	40	E2	60	100	60	100	0	Yes	26 Oct 2026	18 Dec 2026
C2	Construction C2	40	C1	100	140	100	140	0	Yes	4 Jan 2027	26 Feb 2027
T1	Transition T1	25	C2	140	165	140	165	0	Yes	1 Mar 2027	2 Apr 2027
C3	Construction C3	60	T1	165	225	165	225	0	Yes	5 Apr 2027	25 Jun 2027
T2	Transition T2	25	C3	225	250	225	250	0	Yes	28 Jun 2027	30 Jul 2027
P1	Pilot year P1	260	T2	250	510	250	510	0	Yes	2 Aug 2027	28 Jul 2028
C4	Construction C4	120	T2	250	370	365	485	115	No	2 Aug 2027	14 Jan 2028
T3	Transition T3	20	C4	370	390	485	505	115	No	17 Jan 2028	11 Feb 2028
CL	Close-out	5	T3, P1 (FF)	505	510	505	510	0	Yes	24 Jul 2028	28 Jul 2028

Table 7.2 Milestones and their acceptance decisions

ID	Milestone	Date	After	Acceptance / decision
M1	LCO: Lifecycle Objectives	2 Sep 2026	2.4	Vision approved by the Owner; Business Case, charter, Development Case and plans accepted; go for Elaboration
M2	LCA: Lifecycle Architecture, SRS baseline	23 Oct 2026	3.7	SRS signed off by the Owner; architecture, prototype and test plan accepted; gateway and LINE prerequisites verified; financial analysis re-run; go for Construction
M3	IOC: Initial Operational Capability	26 Feb 2027	4.10	Release 1.0 passes the system and load tests (5 transactions a second); go for the pilot on concert nights

ID	Milestone	Date	After	Acceptance / decision
M4	Release 1.0 Core in operation	2 Apr 2027	5.4	Acceptance tests of the two main use cases passed with the Manager and Front Staff; data migrated; Release 1.0 declared in operation
M5	Release 2.0 Growth in operation	30 Jul 2027	6.7	Acceptance tests of the Release 2.0 features passed; pilot year and KPI measurement begin
M6	Release 2.1 in operation	11 Feb 2028	8.6	La Loy migrated to the productized service; Free-plan sign-up released; hand-over to the support routine
M7	Expansion decision, project closed	28 Jul 2028	1.9	Pilot review against the seven success criteria of the charter; expansion decision of the software house; lessons learned; sign-off

Record 5: Effort estimation for tasks

1. Artifact	Effort estimation (Part 2, Section 4): bottom-up person-hours per work package and engineer, reconciled to the baseline of Milestone 1 (2,632 hours: Year 0 2,092 h, Year 1 540 h; labor 846,600 THB), with three-point estimates for the risky packages and the comparison with the top-down estimate.
2. Member	Chawanakorn Sanannam
3. AI Tool	OpenCode (terminal CLI agent at opencode.ai), model "Big Pickle" (opencode/big-pickle); the baseline of Milestone 1 (2,632 h; Year 0 2,092 h; Year 1 540 h; labor 846,600 THB) was pasted.
4. Initial Prompt	**[character]** You are a project estimator working bottom-up per WBS task. **[request]** Prepare SEATS effort estimates using the source baseline of 2,632 hours, Year 0 2,092 hours, Year 1 540 hours and labor cost 846,600 THB. **[examples]** Allocate tasks provisionally and label new task-level hours as estimates. **[adjustments]** Keep each phase total inside its source share and state the basis of estimates. **[type of output]** A work-package effort table plus a basis-of-estimates note. **[evaluation]** The total must reconcile to 2,632 hours / 846,600 THB with zero variance.
5. Initial Output	Summed role-based hours but did not distinguish the source totals from the task-level estimates and mismatched the C1/C2 distribution against the source phase baseline.
6. Review	Review Board row 5: the source estimate is top-down; detailed effort per task is not fully provided. PMBOK correctness: effort is estimated bottom-up and the basis of estimates must be stated; project consistency required the total to hit the 2,632-hour ceiling. A second output still gave hours per phase group only, which is the Milestone 1 table restated, not a bottom-up estimate.
7. Revision	Rebalanced the work-package hours to exact zero variance with the source baseline, kept Year 0 at 2,092 h and Year 1 at 540 h, and labelled the task-level allocation as provisional. By hand, the hours were then estimated per work package and per engineer (49 packages), the four risky packages (gateway payment, load test, scheduled reminders, multi-tenant model) three-point with the PERT average, and the plan model checks that every phase, engineer and year total equals Milestone 1 and prints the percent difference per phase.
8. Final Output	Part 2, Tables 4.2 and 4.3, as delivered below, and the effort per work package and engineer of Table 4.1 (in Part 2): the three-point estimates and the comparison with the top-down estimate (0 % in total, below 8 % per phase). Review Board decision: Revise; accepted after the revision. **Table 4.2 Three-point estimates and PERT weighted average**

WBS ID	Work package	Optimistic O (h)	Most likely M (h)	Pessimistic P (h)	PERT (O + 4M + P) / 6	Allocated (h)	Allocated vs PERT
4.3	Gateway payment and callback	80	100	150	105.0	104	-1.0 %
4.10	Load test, system test and IOC package	90	115	180	121.7	120	-1.4 %
6.1	Scheduled reminders	80	96	150	102.3	100	-2.3 %
8.1	Multi-tenant data model and tenant isolation	72	92	150	98.3	96	-2.4 %

Table 4.3 Bottom-up versus top-down effort by phase

Phase	Boonnisa T.	Sirapatr C.	Chawanakorn S.	Watayut A.	Bottom-up (h)	Milestone 1 top-down (h)	Difference (h)	Percent difference
Inception (I1, I2)	32	40	32	24	128	128	+0	0.0 %
Elaboration (E1, E2)	64	96	96	64	320	320	+0	0.0 %
Construction Release 1.0 (C1, C2)	128	224	256	192	800	800	+0	0.0 %
Transition Release 1.0 (T1)	40	40	50	60	190	190	+0	0.0 %
Construction Release 2.0 (C3)	68.4706	144	168	120	500.471	504	-3.52941	0.7 %
Transition Release 2.0 (T2)	33.5294	30	40	50	153.529	150	+3.52941	2.3 %
Construction Release 2.1 (C4)	40	116	180	108	444	480	-36	7.8 %
Transition Release 2.1 (T3)	8	12	20	20	60	60	+0	0.0 %
Pilot operation and close-out (P1)	20	4	0	12	36	0	+36	200.0 %
Total	**434**	**706**	**842**	**650**	**2632**	**2632**	**+0**	**0.0 %**

Record 6: WBS decomposition and features by release

1. Artifact	WBS decomposition and features-by-release mapping (Part 2, Sections 2.2 and 3.1): deliverable-oriented work packages in noun form, with every feature F1 to F35 assigned to Release 1.0 Core or Release 2.0 Growth and the productization items to Release 2.1.
2. Member	Chawanakorn Sanannam
3. AI Tool	OpenCode (terminal CLI agent at opencode.ai), model "Big Pickle" (opencode/big-pickle); the feature tables and release boundaries of Milestone 1 were pasted.
4. Initial Prompt	**[character]** You are a requirements specifier building a deliverable-oriented WBS. **[request]** Organize the SEATS features by release (Core, Growth and productization) and decompose them into 100 % deliverable-oriented noun work packages. **[examples]** Keep booking/payment/QR check-in separate from reminders/waitlist/transfers/reports and multi-tenant subscription features. **[adjustments]** Map every feature to exactly one work package (100 % rule) and avoid adding new scope. **[type of output]** A WBS table plus a features-by-release table. **[evaluation]** A unit of work appears once; work packages are nouns and every feature sits in exactly one release.
5. Initial Output	The initial feature list mixed features from different releases and the Construction branches were decomposed by page instead of by feature groups, mixing nouns and verbs.
6. Review	Review Board row 6: Core, Growth and productization features were mixed together, and the WBS did not use deliverable-oriented nouns. PMBOK correctness: the decomposition must be mutually exclusive and collectively exhaustive.
7. Revision	Re-grouped the features by release using the source report only and re-decomposed Construction into noun-form feature-group work packages, verifying every feature appears exactly once. By hand, the feature groups were aligned with the iteration contents of Milestone 1 (C1: rounds and packages, table map and hold, payment, sign-in and accounts; C2: e-ticket and messages, check-in, live view and exceptions, audit, backup and consent; C3: reminders, grace and waitlist, transfer, reports) and the features-by-release table became Table 2.1 of the scope statement.
8. Final Output	Part 2, Table 2.1, as delivered below, and the WBS of Table 3.1 (quoted in Record 3): Release 1.0 contains the core booking, payment and QR check-in features; Release 2.0 the reminders, grace extension, waitlist, transfer and reports; Release 2.1 the multi-tenant plans, billing and onboarding, with one numbering scheme shared by all tables. Review Board decision: Revise; accepted after the revision.

Table 2.1 Product scope by release and role

Release	Role	Features	Characteristics
1.0 Core	Customer	F1 LINE sign-in, F2 table map, F3 table selection and 15-minute hold, F4 gateway payment, F5 QR e-ticket, F6 transactional messages, F7 booking terms, F8 my bookings, F9 round list	Self-service booking inside LINE: browse, choose, hold, pay the full fee, keep the e-ticket; no Manager reply needed
1.0 Core	Manager (Owner read-only)	F10 concert rounds, F11 zones and packages, F12 live booking view, F13 ticket escalation management, F14 manual booking and payment overrides	Set up each concert round and watch every table, hold, payment and check-in on one live screen; exceptions by hand, including the degraded payment mode
1.0 Core	Front Staff	F15 QR scan, F16 ticket details, F17 on-site check-in, F18 invalid-ticket escalation, F19 no-show and table release, F20 name lookup	Check-in by QR within 15 seconds per table; release of no-show tables after the grace period
1.0 Core	System	F21 hold release, F22 daily backup, F23 payment callback, F24 staff accounts, F25 audit trail, F26 push-message budget, F27 degraded-mode switch, F28 consent and profile access	First-lock-wins hold with timeout, automatic confirmation on the gateway's webhook, role-based access, audit trail, backup with Excel export, consent
2.0 Growth	Customer and System	F29 grace extension, F30 waitlist, F31 booking transfer, F32 scheduled reminders, F34 reminder scheduling, F35 waitlist offer	Use the central record to lose fewer tables: reminders, grace extension, waitlist in queue order, official transfer
2.0 Growth	Manager and Owner	F33 reports	Booking, revenue, artist-popularity and customer-history reports with export
2.1 Product	Software house	Multi-tenant data model and tenant isolation; Free, Standard and Premium plans with limits and enforcement; subscription billing and setup-fee invoicing; self-service onboarding; service administration; migration of La Loy	The pilot system productized as a subscription service within the 1,000 THB a month ceiling (Section 3.1 of Milestone 1)

Record 7: Consistency review of the drafts, quality requirements and verification

1. Artifact	Consistency review of the three Milestone 2 drafts (decision log, final deliverable, GenAI log) against Milestone 1, the brief and each other, and the quality requirements with their verification (Part 2, Table 2.2 and the evidence column of Table 2.4): every target of Table 2.19 of Milestone 1 linked to a test, a threshold and the work package that runs it.
2. Member	Watayut Aiamanan
3. AI Tool	Claude Code (Anthropic, model Claude Fable 5.1), run inside the project repository, with the Milestone 1 report sources, the project brief, the teammates' three drafts and the financial model (tools/finance.py) readable by the agent.
4. Initial Prompt	**[role]** Claude Code working in the 2110722 repository, where the Milestone 1 report, its build tools and the brief are the context. **[task]** "Please check the teammates' Milestone 2 drafts (workspace/received/examples/teammate_milestone_2) and merge them with the current Milestone 1"; then, before merging: "please check any inconsistency in the M2 draft also; if you find one, let me know." **[format]** The report sources of the Milestone 1 pipeline (markdown built to PDF) and, first, a list of the inconsistencies found.
5. Initial Output	A list of inconsistencies in three groups. Conflicts with Milestone 1: the drafts named the system SEATS while Milestone 1 said CTRS; the scope statement used requirement identifiers R1 to R7 that exist in no project document; the critical-path table's early-start dates ignored the year-end break and the gaps between iterations (C2 on 21 December instead of 4 January, T1 on 15 February instead of 1 March); pilot-monitoring hours were folded into the Release 2.1 pool. Internal contradictions: the decision log named the productization branch as the zero-slack path while the deliverable gave it 121 days of float; verb labels remained although the log recorded their replacement by nouns; a task started before its predecessor finished; summary durations did not equal their children; three-point estimates were claimed but no values shown; three decisions read Accept in the log and "Accept with edits" in the records. Gaps against the brief: effort per task, assignment of resources to tasks and a cost derived from them were missing, the Gantt and critical path were at iteration level, and the quality requirements had thresholds but no verification method. The agent also proposed the quality-requirements table with a verification, a threshold and a work package per target.
6. Review	Requirement satisfaction: the review covered the three criteria of the brief for every artifact of the drafts, and the quality table answered the Review Board's row 7 for all seven targets (six of Table 2.19 plus the operating-cost ceiling of the charter). PMBOK correctness: each verification is a test with a measurable threshold run by the Quality Assurance role, not by the author of the function, and the date findings used the working-day logic of the critical path method. Project consistency: every finding was checked against the Milestone 1 tables it cited (2.13 to 2.21, 3.4, 3.6, 4.1) and found correct,

	including the name of the project, which the team had in fact changed to SEATS after Milestone 1 without recording it.
7. Revision	The member answered the agent's questions with the team's decisions: the project name is SEATS (recorded in the Document Revision History), the eight rows of the decision log stay as the teammates wrote them with the section references re-pointed, the prompt records of the other members are aligned but not rewritten, and the merged deliverable goes to task level. The agent then merged the drafts into the Milestone 1 format, replaced R1 to R7 by the feature identifiers, placed each verification of the quality table in the work package and iteration that runs it (4.10, 5.1, 5.4, 8.5, 1.8), and built the remaining tables from a plan model (Record 8).
8. Final Output	Part 2, Table 2.2, as delivered below, and the evidence column of Table 2.4 (quoted in Record 2); the findings of the review are resolved in Part 2 as follows: one name (SEATS), feature identifiers F1 to F37, dates from the working-day calendar of Milestone 1 (Tables 3.2 and 7.1), one critical path justified by the float table (Section 7.2), noun work packages (Table 3.1), three-point values shown (Table 4.2), pilot KPI work and close-out as work packages of their own (1.8, 1.9). Acceptable because every quality target has a measurable threshold and an independent test, and because no finding of the review remains open. Review Board decision: Revise; accepted after the revision.

Table 2.2 Quality requirements and their verification

Quality requirement	Target (Table 2.19 of Milestone 1)	Verification (test evidence)	When
Throughput at booking-open time	at least 5 transactions a second	Load test with simulated customers opening a round: 5 transactions a second sustained for 10 minutes, one hold per table, no deadlock (WBS 4.10)	before IOC; repeated for Release 2.1 (8.5)
Response time	within 2 seconds for customer and Front Staff transactions	Response-time measurement during the load test and in the acceptance tests on the test devices inside LINE (4.10, 5.1)	IOC, T1
Check-in time	within 15 seconds per table	Timed check-in of 30 tickets by the Front Staff with the tablet at a pilot night (5.1, 5.4)	T1
Recovery after an outage	within 2 hours during opening hours	Restore drill: restore the daily backup to a fresh environment and bring the service up within 2 hours (4.10, repeated in 8.5)	IOC, C4
Backup and export	daily backup of booking data with export to Excel	Backup job verified daily in the test environment; the Excel export opened by the Manager (4.9, 5.1)	C2, T1
Supported browsers	current LINE in-app browser; last two versions of Chrome, Safari and Edge	Browser-compatibility checklist run on the test devices and the shop's desktop browser (4.10, 5.1)	IOC, T1
Operating cost of the restaurant	hosting, APIs and LINE plan at or below 1,000 THB a month	Monthly cost review of hosting and push messages during the pilot year (1.8)	P1

Record 8: Cost estimate consistency and the plan model

1. Artifact	Cost estimate consistency (Part 2, Section 6) and the plan model behind Sections 3 to 7: one source (tools/plan.py) that holds the 49 work packages with their hours per engineer, dates, predecessors, owners, three-point estimates and other costs, and generates the WBS, activity, effort, RACI, loading, cost, time-phased baseline, CPM and critical-path tables and the Gantt chart on every build, checking the totals against Milestone 1.
2. Member	Watayut Aiamanan
3. AI Tool	Claude Code (Anthropic, model Claude Fable 5.1), run inside the project repository (same session as Record 7).
4. Initial Prompt	**[role]** Claude Code in the repository, with the Milestone 1 effort and cost tables (Tables 2.20, 3.4 to 3.7) and the financial model as context. **[task]** Merge the drafts at full task level: effort per work package and engineer, resources assigned to the work packages with rates, a cost estimate derived from them with the contingency reserve, and the schedule with Gantt chart and critical path, all reconciled with Milestone 1. **[format]** Generated tables in the report sources, regenerated on every build, with a check that the totals equal Milestone 1.
5. Initial Output	The agent reported that the draft's cost table copied the Milestone 1 figures by category and could not show that they follow from the resource assignment, the rates and the effort, that the contingency was a bare 10 % without the risks it covers, and that the management reserve was mentioned but not decided. It then produced the plan model: 49 work packages whose hours per engineer sum to the phase totals of Table 3.4 of Milestone 1, the other costs of Tables 3.5 and 3.6 attached to the work packages that use them, labor as hours times rate, contingency per year, a time-phased baseline by iteration, the forward and backward pass on the iteration network with a working-day calendar, and the Gantt chart. Its checks printed the totals per engineer, per year and per phase and the weekly load of each engineer, which showed peaks above the Milestone 1 availability in the middle of the Construction iterations.
6. Review	Requirement satisfaction: items 4 to 7 of the brief are now derived from one model, and the cost reconciles to 1,010,590 THB, 767,690 THB in Year 0 and 242,900 THB in Year 1, labor 846,600 THB for 2,632 hours, net investment 860,590 THB. PMBOK correctness: the budget build-up (work-package estimates, contingency reserve, cost baseline, management reserve, project budget) is explicit, the contingency reserve is in the baseline and tied to identified risks, effort and duration are kept apart, and the critical path is justified by total slack. Project consistency: the model refuses to build if a total differs from Milestone 1. Weaknesses: two decisions had been written by the agent on its own (no management reserve; the per-risk split of the Year 0 reserve), and the loading peaks needed a stated leveling policy.

7. Revision	The member confirmed the two decisions as the team's (no management reserve, so that the project budget equals the cost baseline of the charter; the per-risk split kept as provisional amounts until the Milestone 3 risk register) and asked for the loading to be shown honestly: the agent moved the test plan (3.6) and the Release 2.1 test (8.5) earlier in their iterations and wrote the leveling policy of Section 5.3 (planned averages equal the availability; peaks leveled by the engineer sequencing overlapping packages inside their free slack; milestone dates unchanged). Arithmetic was not done by the model's prose but by the code, which is kept with the report sources.
8. Final Output	Part 2, Table 6.1, as delivered below (the time-phased baseline, Table 6.2, and the reserve rules are in Part 2; the effort and schedule tables of the same model are quoted in Records 3, 5 and 4). Acceptable because every figure of the cost baseline is traceable to hours, rates and listed items, the reserves are decided and justified, and the reconciliation with Milestone 1 is exact. Review Board decision: Accept.

Table 6.1 Cost estimate by WBS item, contingency reserve, cost baseline and project budget

WBS	Item	Hours	Labor basis (hours x THB/h)	Labor (THB)	Other costs (THB)	Subtotal (THB)
1	Project Management	246	Boonnisa T. 222 h x 350; Sirapatr C. 4 h x 350; Watayut A. 20 h x 300	85,100	-	85,100
2	Inception (I1, I2)	96	Boonnisa T. 8 h x 350; Sirapatr C. 40 h x 350; Chawanakorn S. 32 h x 300; Watayut A. 16 h x 300	31,200	Development and staging hosting, August 2026 300	31,500
3	Elaboration (E1, E2)	292	Boonnisa T. 36 h x 350; Sirapatr C. 96 h x 350; Chawanakorn S. 96 h x 300; Watayut A. 64 h x 300	94,200	Development and staging hosting, September to October 2026 600	94,800
4	Release 1.0 Core: Construction (C1, C2)	736	Boonnisa T. 64 h x 350; Sirapatr C. 224 h x 350; Chawanakorn S. 256 h x 300; Watayut A. 192 h x 300	235,200	Test devices: Android phone and 10-inch tablet 15,000; Development and staging hosting, November 2026 to February 2027 1,200	251,400
5	Release 1.0 Core: Transition (T1)	178	Boonnisa T. 28 h x 350; Sirapatr C. 40 h x 350; Chawanakorn S. 50 h x 300; Watayut A. 60 h x 300	56,800	Domain name and TLS certificate 1,200; Development and staging hosting, March 2027 300; Production hosting of the pilot restaurant, March 2027 700	59,000
6	Release 2.0 Growth (C3, T2)	608	Boonnisa T. 56 h x 350; Sirapatr C. 174 h x 350; Chawanakorn S. 208 h x 300; Watayut A. 170 h x 300	193,900	Development and staging hosting, April to July 2027 1,200; Production hosting of the pilot restaurant, April to July 2027 2,800	197,900

WBS	Item	Hours	Labor basis (hours x THB/h)	Labor (THB)	Other costs (THB)	Subtotal (THB)
7	Pilot operation (P1)	0		0	Production hosting during the pilot year 8,400; support labor 96 h x 300 = 28,800	37,200
8	Release 2.1 Product (C4, T3)	476	Boonnisa T. 20 h x 350; Sirapatr C. 128 h x 350; Chawanakorn S. 200 h x 300; Watayut A. 128 h x 300	150,200	Company registration, terms of service, privacy policy and data-processing agreement 15,000	165,200
	Work-package estimates	**2632**		**846,600**	**75,500**	**922,100**
	Contingency reserve, Year 0 (RISK-02, RISK-03, RISK-05, RISK-08, RISK-10)		10 % of 697,900 THB			69,790
	Contingency reserve, Year 1 (RISK-04, RISK-07, RISK-12)		10 % of 187,000 THB (labor of C4 and T3 and the legal set-up)			18,700
	Cost baseline					**1,010,590**
	Management reserve		none held by the sponsor for this project (unknown unknowns escalate to the partners' meeting)			0
	Project budget (charter)					**1,010,590**
	Less: pilot implementation fee of La Loy (revenue inside the project)					(150,000)
	Net investment of the software house					**860,590**